

Over the past 75 years the study of kaon physics has been integral to the development of the standard model. Since 2016, the NA62 experiment has been the leading experiment in the field of kaon physics. However, with the NA62 experiment scheduled up to CERN Long Shutdown 3 (LS3), which is expected in autumn 2026, it is important to investigate future experiments. It was then required to investigate the future of kaon physics and the potential experiments that could succeed NA62. This chapter will discuss the potential future of kaon physics, the experiments that could be used to further the field, and the work I contributed to these experiments.

0.1 High Intensity Kaon Experiments (HIKE)

The High-intensity kaon experiments (HIKE) was a proposed project to continue kaon physics at CERN in the ECN3 experimental hall and represented a long-phased program after LS3. HIKE would have profited from a four to six-fold increase in beam intensity compared to NA62 and included cutting-edge detector technologies that will be described in this section. The beam intensity requirements for HIKE are described in Table 1. To accommodate the higher intensities, upgrades were needed to the infrastructure that provided the beam to ECN3. A number of work groups such as the ECN3 beam delivery group within the Physics Beyond Colliders community at CERN [1] was already conducting efforts to consolidate all the work required to upgrade infrastructure. Three phases were foreseen within the HIKE program, the first being a continuation of NA62 charged kaon physics with the high-intensity beam and detector upgrades to accommodate the rate increase with the goal of measuring the branching ratio of $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ to 5 % precision. The second phase would have involved neutral kaon physics with charged secondary particle tracking with the goal to measure the branching ratio of $K_L \rightarrow \pi^0 l^+ l^-$. The third phase would have been designed to observe the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay and measure its branching ratio to a 20 % precision.

HIKE Phase	Protons on Target/Spill	K Decays/Year
Phase 1 (K^+)	1.3×10^{13}	2×10^{13}
Phase 2 (K_L + tracking)	2×10^{13}	3.8×10^{13}
Phase 3 (K_L , KLEVER)	2×10^{13}	1.3×10^{13}

Table 1: Intensity requirements for three phases for HIKE. The number of kaon decays per year corresponds to those that decay within the HIKE fiducial volume.

0.1.1 Phase 1

The experimental design of NA62 has been proven as a successful setup for the precision measurement of the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay, described in Section ?? . The HIKE Phase 1 detector setup shown in Figure 1, is similar to the NA62 setup described in Section ?? with new and upgraded detectors. The HIKE detectors were required to withstand a four times greater particle rate with respect to NA62 equating to 1.3×10^{13} protons per spill on the target. The expected beam had similar characteristics to NA62 with a 75 GeV/c secondary beam produced from the 400 GeV/c primary beam from the SPS accelerator onto a target at an angle of 0 rad.

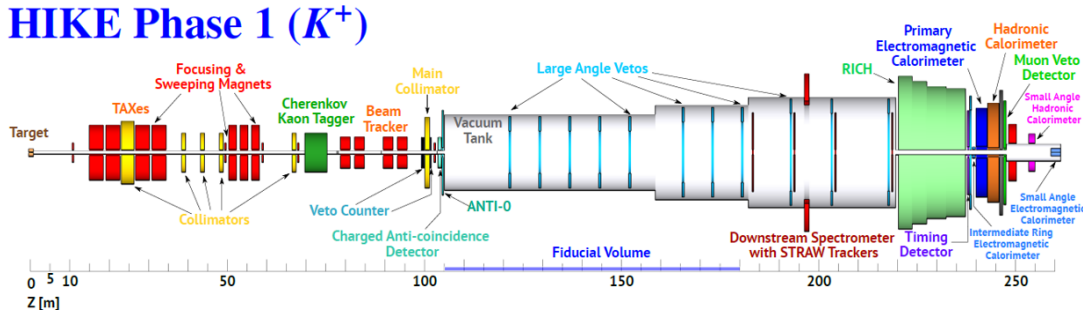


Figure 1: Diagram of HIKE Phase 1 detector setup [2].

The detectors before the fiducial volume included an upgraded photon-detection system for the KTAG, described in Section 0.2, and a new GTK chip design. It was proposed to upgrade the existing NA62 Vetocounter with scintillating fibre (SciFi) technologies to improve the granularity and to accommodate the increased beam intensity.

Major upgrades were envisioned to the LAV and STRAW detectors within the

fiducial volume. A number of LAV stations were to be upgraded to a new lead scintillator design, similar to the Vacuum Veto System (VVS) planned for the CKM experiment [3, 4]. The LAV detectors were crucial to the subsequent phases of HIKE, especially Phase 3, and were designed to fulfil the requirements of the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ measurement even when used in Phase 1. The STRAW spectrometer was to be upgraded to use novel 5 mm straw design with a layout shown in Figure 2. The reduction in the straw diameter would have decreased drift times while increasing granularity and resulting in an expected improvement in the track angle and momentum resolution of 10-20 % compared to NA62.

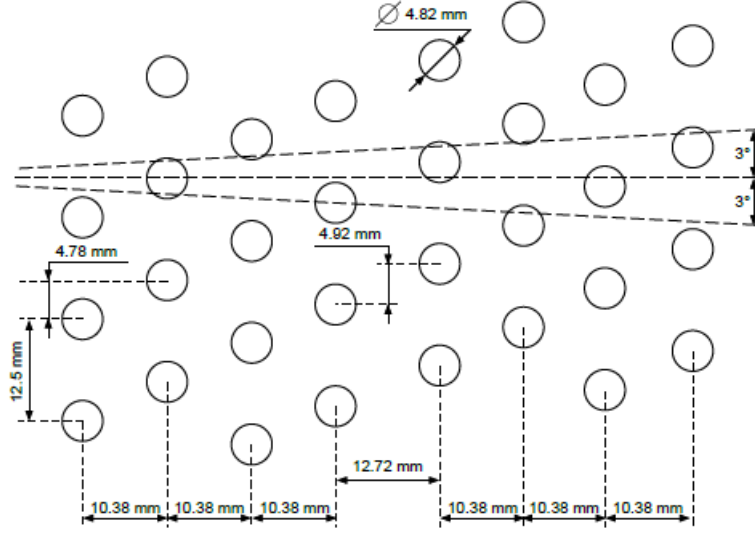


Figure 2: Diagram of the new STRAW spectrometer layout with 5 mm straw tubes [2].

Downstream of the fiducial volume, the PMT arrays of the RICH detector would have been upgraded, described in Section 0.3. The HIKE timing detector would profit from the current NA62-CHOD detector with upgraded PMTs, which were to be chosen from the outcome of studies from other detectors such as KTAG, RICH and MUVs. For HIKE Phase 1, the LKr was proposed as the primary electromagnetic calorimeter given its excellent energy resolution, but the required timing resolution was to be achieved by upgrading its readout system. A number of other detectors would have been upgraded or replaced with higher granularity, such as the hadronic calorimeter, MUVs, IRC, and SAC.

It was expected that Phase 1 would run between LS3 and LS4, with the exact

years being undetermined. About 2.5×10^{-6} kaon decays in the fiducial volume was expected per proton on target during HIKE phase 1. Given the expected protons per spill of 1.3×10^{13} (four times NA62) and a nominal year of 200 data-taking days with an average of 3000 spills per day, would result in an expected 2×10^{13} kaon decays per year were foreseen during HIKE phase 1. As a consequence the Phase 1 goals could be completed in 5 years of data taking.

0.1.2 Phase 2

HIKE Phase 2 would have been a multi-purpose neutral K_L decay experiment with the primary goal being to measure the branching ratios of the $K_L \rightarrow \pi^0 l^+ l^-$ decays. The detector setup for phase 2, shown in Figure 3, would have involved minimal modification to the experimental setup of Phase 1, with the removal of detectors that were only applicable to the charged kaon beam. The KTAG, GTK, RICH, and SAC detectors would have been removed, and the STRAW spectrometer would have been shortened to account for a softer momentum of the secondary particles. The STRAW chambers would have been reconfigured to have central beam holes to account for the neutral beam. The MNP33 magnet required a reduction in strength by 20% to account for the shorter spectrometer, resulting in a momentum kick of 210 MeV/c. The fiducial volume would have been increased to 90 m compared to 75 m in Phase 1.

HIKE Phase 2 (K_L)

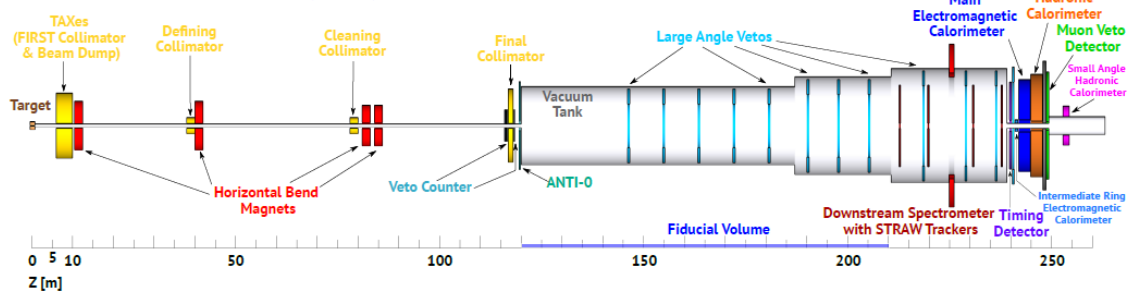


Figure 3: Diagram of HIKE Phase 2 detector setup [2].

The design proposed a production angle of 2.4 mrad on the target producing an expected K_L yield of 5.4×10^{-5} per proton. Assuming an integrated proton flux of 1.2×10^{19} protons on target per year would have resulted in 3.8×10^{13} K_L decays

in the fiducial volume per year. The mean beam momentum at the entry of the fiducial volume was expected to be 79 GeV/c with a mean beam momentum for K_L which decay of 46 GeV/c.

HIKE Phase 2 was planned to follow after Phase 1, which in turn was scheduled after LS4 up to LS5. The expected protons per spill in Phase 2 was six times that of NA62 at 2×10^{13} . Assuming 200 days of data taking and 3000 spills per day (as in Phase 1) and an 3.2×10^6 K_L per proton on target would result in 3.8×10^{13} decays per year. The expected time to Phase 2 goals was 6 years of data taking.

0.1.3 Phase 3 (KLEVER)

HIKE phase 3, previously known as KLEVER [5], was an experiment designed to measure the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay. The experimental setup is shown in Figure 4 and differs from the design used at NA62 and HIKE Phase1/2. The setup consists of a large number of photon detectors located around a 160 m long vacuum volume to produce a complete coverage of photons produced from K_L decays. The detectors include coverage for photons emitted up to 100 mrad and allow for photons emitted with an angle up to 7.5 mrad to be detected by the main electromagnetic calorimeter (MEC).

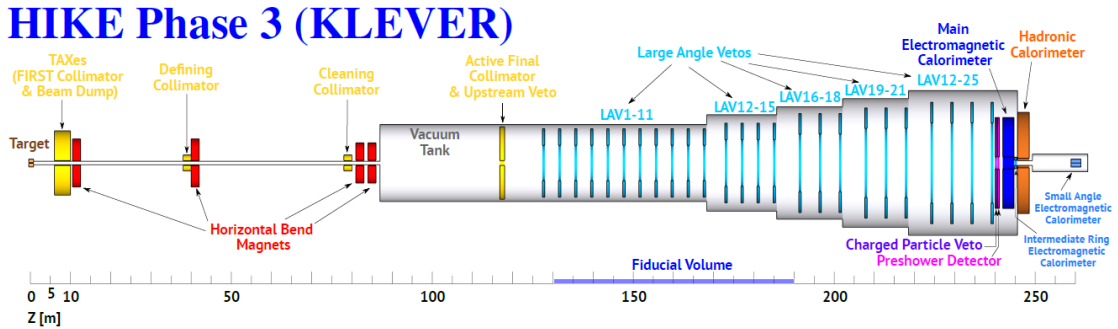


Figure 4: Diagram of HIKE phase 3 detector setup [2].

Due to the significantly different experimental setup, Phase 3 was separated from the HIKE proposal submitted in 2023 [2]. However, some detectors would have been shared between the two experiments as the granularity and efficiency requirements for Phase 3 were driven the detector requirements for the first two phases, reducing the overall costs. The LAVs, MEC, IRC, SAC and the hadronic calorimeters, which would have been built for HIKE Phase 1/2, had all designs driven by their usage

in Phase 3. The upstream veto, charged particle veto, and the pre-shower detectors are unique to Phase 3.

0.2 The KTAG upgrades for HIKE

To sustain the increased kaon rate, the KTAG required upgrades to the photon-detection and data acquisition systems, allowing to reach the required time resolution of 15-20 ps and kaon identification efficiency of greater than 95 %. The expected kaon rate in HIKE Phase 1 was 200 MHz, which corresponds to a maximum rate of 10 MHz/cm² for the detected photons at the PMT array. This rate takes into account the production of Cherenkov photons in CEDAR-H and the resultant acceptance of the photons on the PMT plane. It was proposed that the NA62 PMT arrays were replaced with arrays of micro-channel plate photomultipliers (MCP-PMTs). The MCP-PMTs would provide a single photon time resolution of 50-70 ps depending on the model, a low dark rate of 1 kHz/cm² and the ability to sustain the required photon rate. A downside of the MCP-PMTs is their limited lifetime from the ageing of the photocathode caused by feedback ions from the residual gas resulting in a rapid reduction in the Quantum Efficiency (QE) [6]. This results in the rapid degradation of PMT quantum efficiency with the increasing integrated anode charge. MCP-PMTs are similar to normal PMTs except that the dynode section is replaced by a micro-channel plate. An example diagram is shown in Figure 5, where a photon hits the photocathode window, producing a single electron, which is then accelerated towards the MCP by an electric field provided by a voltage difference across the gap. The electron then collides with the walls of the MCP, producing secondary electrons, which results in a cascade collected by the anodes below. The dual layer design, or chevron, adds additional benefits compared to a single layer design such as better gains due to the increased surface for secondary electrons to be produced

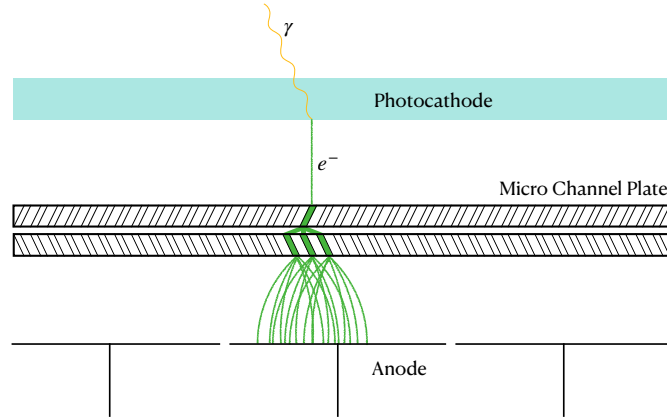


Figure 5: Diagram describing the workings of a MCP-PMT.

Major breakthroughs to improve the lifetime of the MCP-PMT have come with the method of Atomic Layer Deposition (ALD) [7], where the channels of the MCP are coated with a ultra-thin layer of Al_2O_3 or MgO . The layer prevents the out-gassing from the MCP glass substrate and drastically reduces the heavy ions reaching the photocathode resulting in significant increases in lifetime of the MCP-PMTs [8]. Utilising the method of ALD, MCP-PMTs from Photonis have been demonstrated to withstand an Integrated Anode Charge (IAC) greater than 20 C/cm^2 [9], corresponding to a lifetime improvement of a factor of 100 compared to standard MCP-PMTs, shown in Figure 6.

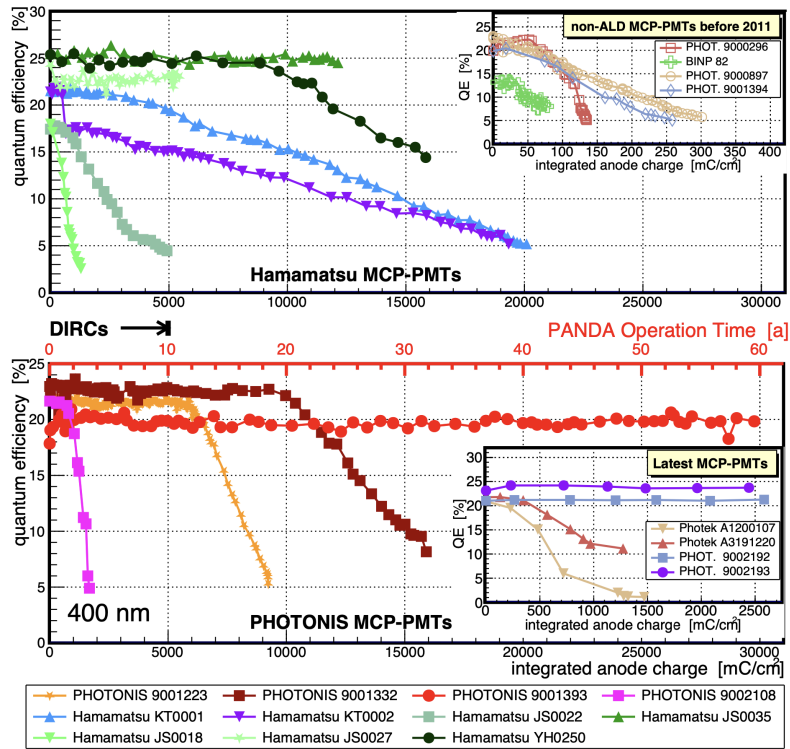


Figure 6: Lifetimes of available MCP-PMTs with the photocathode QE vs Integrated Anode Charge (IAC) for photons at 400 nm wavelength. Top, old ALD coated MCP-PMTs from Hamamatsu with non coated versions shown in the insert plot. Bottom, more recent ALD coated MCP-PMTs from PHOTONIS and Photek in the inserted plot. [10]

An example of MCP-PMT model proposed for the KTAG is shown in Figure 7. The module consists of an 8x8 array of pixels which can individually be read out.

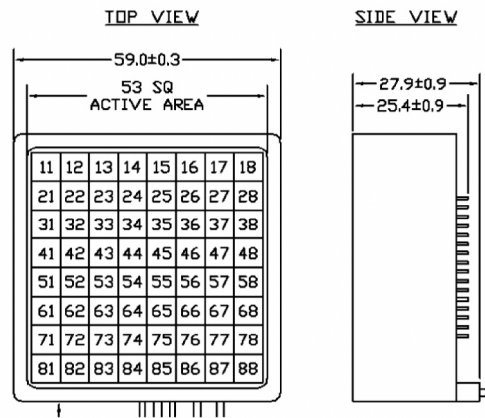


Figure 7: Schematic of Photonis MCP-PMT [11].

Simulations of the KTAG with the upgraded photon-detection system for use in HIKE Phase 1 show that each of the eight KTAG sectors would use four of these MCP-PMT, shown in Figure 8. The four MCP-PMTs per sector were required for full coverage of the light spot produced when the light exiting the CEDAR was reflected by the spherical mirror. The schematic in Figure 7 shows there was some unused space around the active area of each MCP-PMT, which would result in reduced efficiency, especially in the centre, where the maximum amount of light would be expected. To mitigate this, reflective surfaces were attached to each side of the individual MCP-PMTs shown in Figure 8 to reflect the light onto the active surface.



Figure 8: Visualisation of the MCP-PMT array placed inside of the KTAG

The resultant light spot for one of the sectors is shown in Figure 9, where the inactive areas between the individual MCP-PMT and the increased amount of light at the edges produced from the reflective surfaces are shown. The position on the MCP-PMT in the KTAG sector was optimised to produce the maximum number of photo-electrons per kaon candidate.

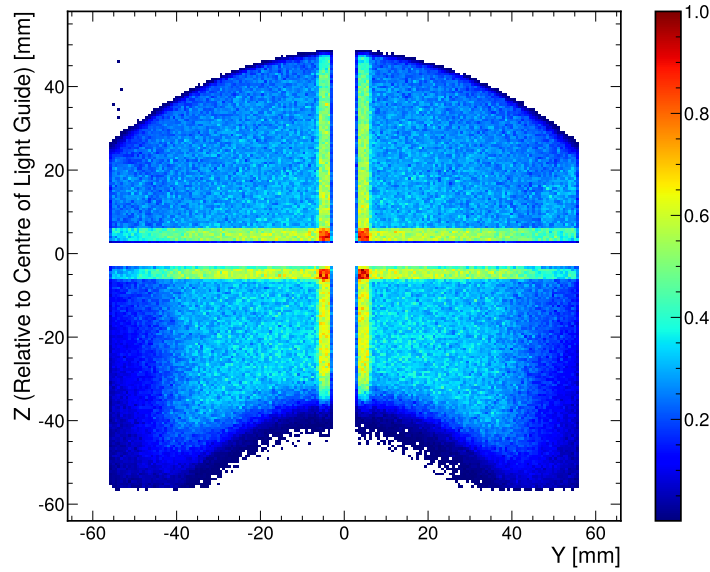


Figure 9: Collected light distribution for the MCP-PMT installed in the KTAG.

HIKE Phase 1 required that the MCP-PMTs installed within the KTAG photon-detection system to have a demonstrated lifetime of 80 C/cm^2 . Other requirements for HIKE for the MCP-PMT is a single photon detection rate of 10 MHz/cm^2 without degradation to the signal due to charge saturation effects. The MCP-PMTs manufactured by Photonics with 2 layers of ALD provide the best lifetime with no degradation in QE after 30 C/cm^2 , shown in Figure 6. An excellent candidate for MCP-PMT was Photonis XP85012-S-R2D2 of the PLANACON line. However, the Photonis MCP-PMT doesn't match with the HIKE Phase 1 rate requirements and result in reduced gain above 1 MHz/cm^2 , shown in Figure 10. It is possible to reach 10 MHz/cm^2 with smaller sensor size but would result in a quadrupling of the cost to cover the light spot. Future research looking at reducing the resistance, capacitance and gain for single photon detection could possibly allow the larger XP85012-S-R2D2 to accommodate up to 10 MHz/cm^2 .

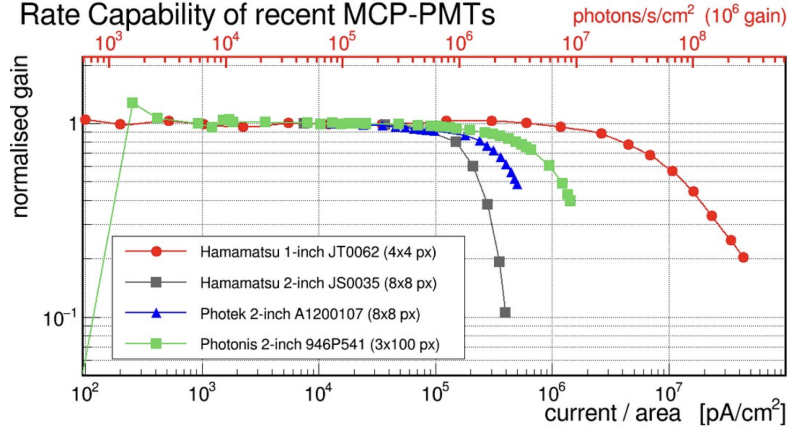


Figure 10: Rate capability of state of the art MCP-PMTs [12].

The Photonis MCP-PMT has a Collection Efficiency (CE) of above 60 %, the probability of the photo-electron passing through the pores of the MCP and producing a signal. With current technologies it is possible for MCP-PMTs reach CE of over to 90 % [13], however the Photonis MCP-PMT was selected as it better fit the HIKE Phase 1 timing requirements requirements as the higher CE models utilise late photo-electrons resulting in higher time resolutions. The XP85012-S-R2D2 MCP-PMT had a transit time spread of 50 ps with a timing precision RMS of 120 ps [14] and results in a time response, shown in Figure 11 which was implemented used for the HIKE simulations.

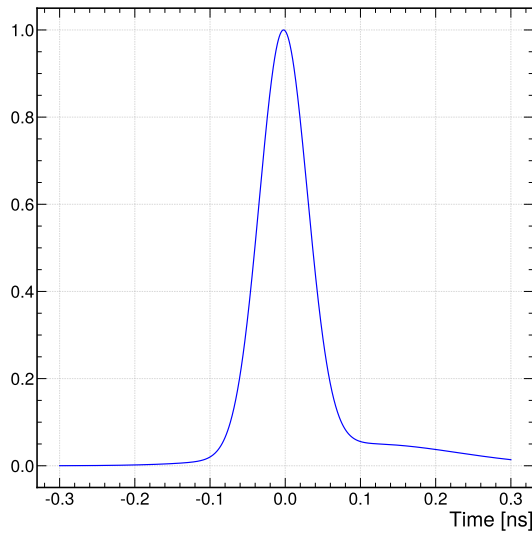


Figure 11: Time response of the Photonics XP85012-S-R2D2 MCP-PMT.

The QE for the MCP-PMT from Photonis is shown in Figure 12. There are

two options for the photocathode of the MCP-PMT available: a standard Bialkali (Sb-Rb-Cs, Sb-K-Cs) or a Hi-QE Blue. The HI-QE are more recent models, however there is no lifetime data available so albeit being the preferred option the standard Bialkali was chosen for use in the simulation. Further studies at The University of Birmingham are planned to characterise the Photonis XP85012-S-R2D2 MCP-PMT by measuring the lifetimes, QE, CE, and other characteristics in order to decide which type would have been used for high-intensity kaon physics at HIKE.

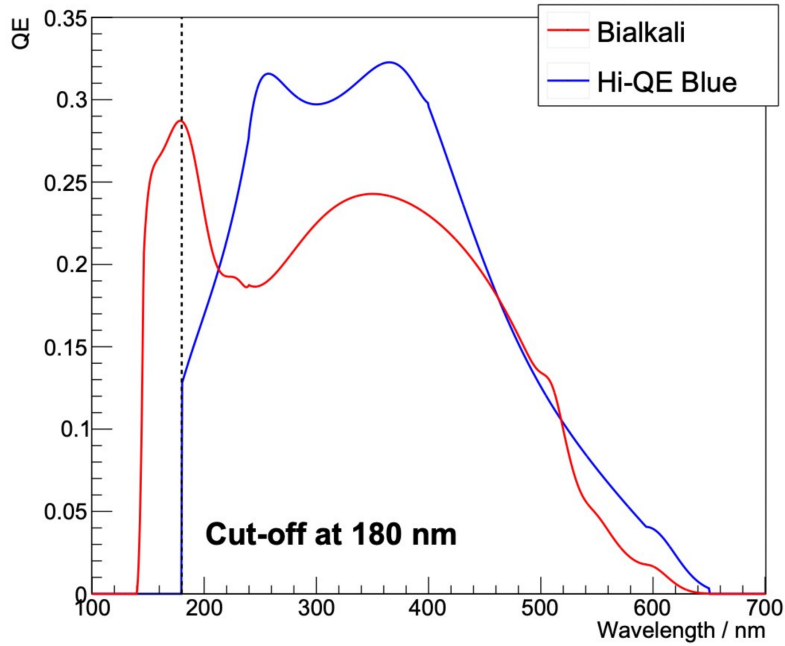


Figure 12: Quantum Efficiency of Photonics MCP-PMT as a function of wavelength.

From the Monte Carlo simulation, the implemented MCP-PMTs performed excellently with the result shown in Figure 13. The MCP-PMT produced 24.3 photoelectrons per kaon candidate with a time resolution of 18 ps, taken from the Gaussian sigma of the distribution in Figure 13 (Right). The results surpassed the requirements for the KTAG in HIKE and the upgrades were ready to be used in the HIKE simulations.

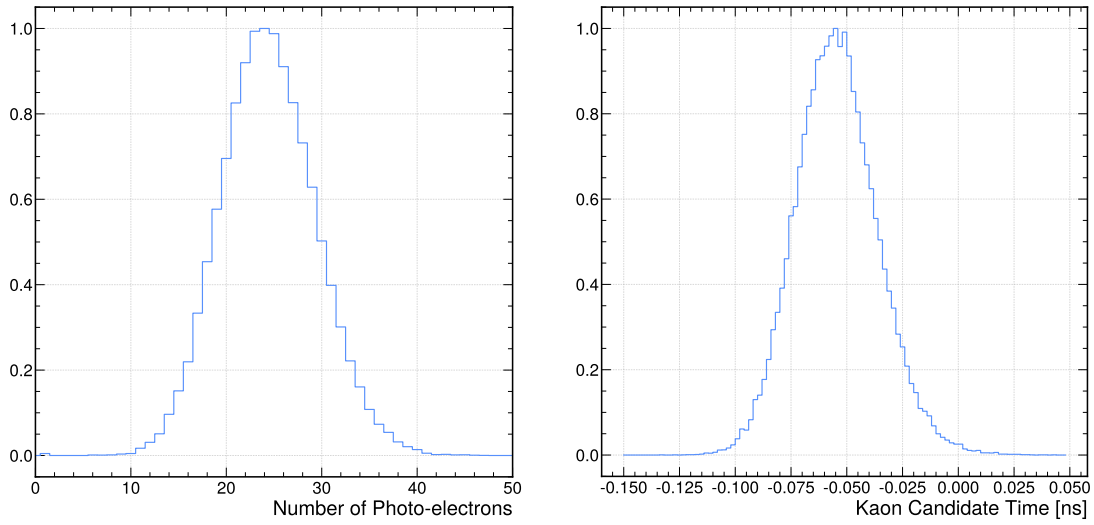


Figure 13: Results from the Monte Carlo simulation of CEDAR-H and MCP-PMT. Left, the number of photo-electrons per kaon candidate and right, the Kaon Candidate time.

0.3 RICH Upgrade

The overall design of the NA62 RICH described in section ?? was well suited for re-use within HIKE with an upgrade to the photon-detection system. The NA62 RICH photon detection system comprises two arrays on either side of the beam pipe, equipped with Hamamatsu R7400-U03 PMTs placed in a pattern described in Figure 14. The PMTs have a single hit time resolution of 240 ps and a quantum efficiency of 20 %. Due to the placement of the PMTs, the distance between the centre of each PMT is 18 mm, which is constrained by the sensor size and the geometry of the PMT itself. This distance between sensors was the main contributor to the single hit position resolution of 4.7 mm, resulting in an overall ring radius resolution of 1.5 mm. The number of hits per ring candidate and the single hit time resolution produced a time resolution of 80 ps. Due to the increased beam intensity, the main requirement of the RICH for HIKE was a time resolution of 20 - 30 ps.

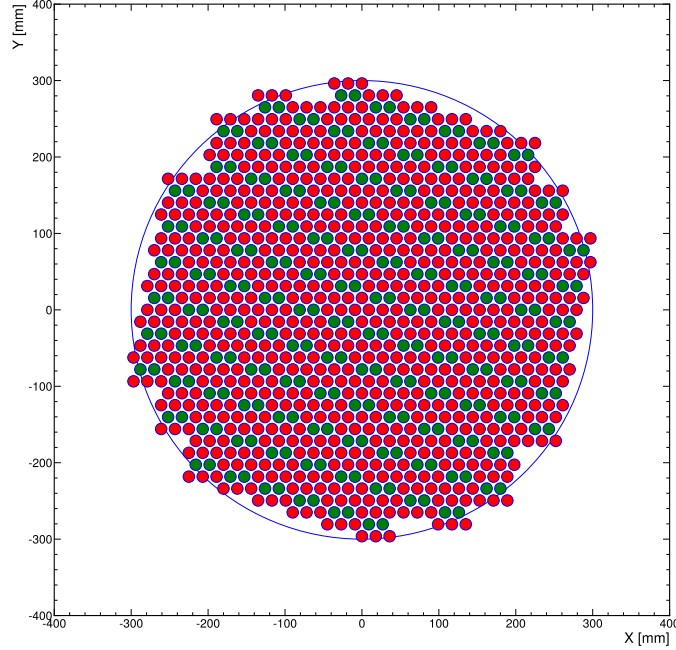


Figure 14: Diagram of RICH PMT array

The proposed RICH photon detection system upgrade would replace the current PMTs with a smaller sensor size, allowing for a smaller distance between sensor centres and reducing the hit position resolution. An excellent candidate for a sensor replacement was silicon photomultipliers (SiPMs). SiPMs with small sensor sizes, high quantum efficiency of 40 % and a single hit time resolution of at least 100 ps are available from major manufacturers. Three sensor sizes were selected for further investigation, $9 \times 9 \text{ mm}$, $6 \times 6 \text{ mm}$, and $3 \times 3 \text{ mm}$. Each SiPM sensor size had apparent benefits and downsides, and a balance between performance and complexity/price had to be taken into account. The smaller sensors would produce a better hit position resolution but at an increased cost and complexity. Whereas the larger sensors would have worse hit position resolution but at reduced complexity and cost. A study was conducted to determine the optimal sensor size for the SiPMs, balancing overall detector performance with cost and complexity.

Using the upgraded NA62 software package described in Section ??, it was possible to implement the new PMTs into the RICH Monte Carlo simulation with the implemented arrays shown in Figure 15. Compared to the PMT arrays currently

used in NA62, the PMs have been aligned on the vertical axis to simplify the placement because of the square geometry of the new PMTs. The area of the expected light spot, shown in Figure 15, as a grey circle, is completely covered by the SiPMs with no gaps between the sensors which was a clear improvement compared to the NA62 setup shown in Figure 14.

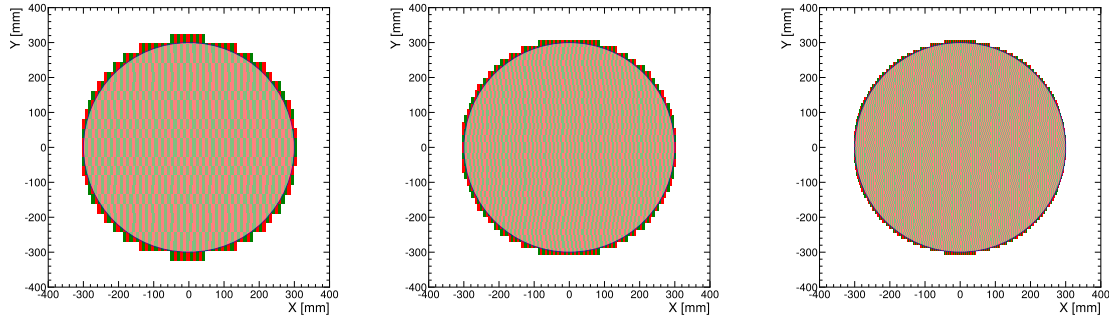


Figure 15: Diagram of proposed RICH SiPM arrays for use in HIKE. Left, 9x9 mm, center 6x6 mm and right 3x3 mm.

The performance for HIKE required a SiPM with a Quantum Efficiency (QE) of at least 40 % [2]. The Hamamatsu S13360-**50CS series was selected as the preliminary candidate for the RICH PMT upgrade to achieve this requirement. Hamamatsu offers a number of SiPMs in the S13360 series, with the **50CS selected for its improved Ultra Violet (UV) sensitivity and high quantum efficiency of 40 % at 460 nm, shown in Figure 16.

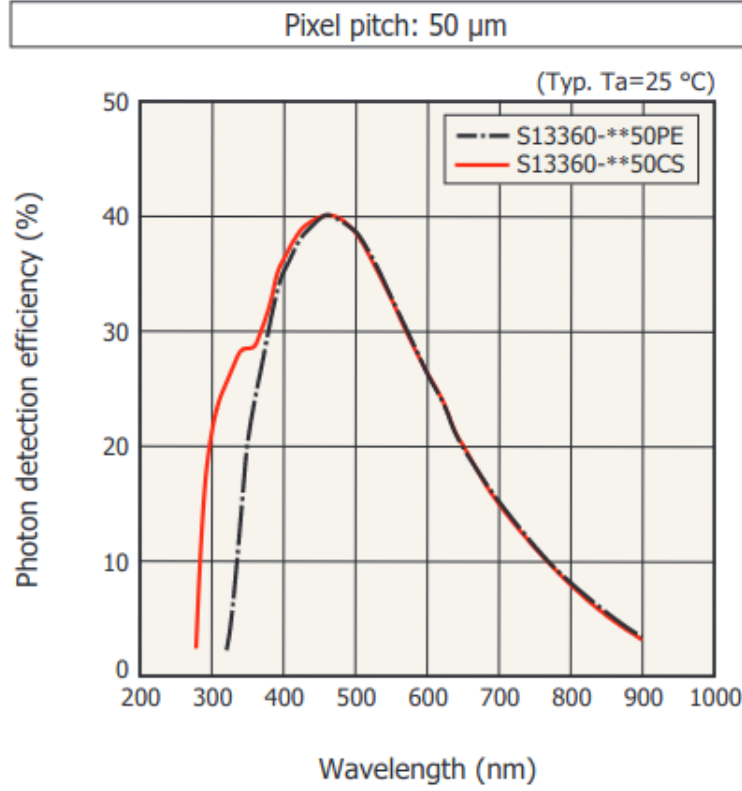


Figure 16: Selected SiPM quantum efficiency taken from Hamamatsu S13360 data-sheet [15]

Along with the QE efficiency, a single hit time resolution of 100 ps was required [2]. The manufacturer stated the Hamamatsu S13360-**50CS series has excellent time resolution, but no definite value was given. The time resolution would also depend on the final design of the RICH readout electronics and would have to be measured in a test setup. The SiPMs from Hamamatsu have previously been tested and have reached a single hit time resolution below 100 ps [16]. For use in the simulation a conservative simple Gaussian time response curve with a sigma of 100 ps was used for the simulations, which would be updated later on for a specific model to be implemented when chosen.

The expected performance of the RICH detector with the new PMTs was compared to the current NA62 RICH detector in Table 2. With the increased quantum efficiency, the expected number hits per π^+ between 15 and 45 GeV/c was expected to increase by a factor of two. This, along with the improved time resolution of the new PMTs resulted in an expected time π^+ time resolution of 27 ps at 15 GeV/c and 20 ps at 45 GeV/c . Using the implemented upgrades to the RICH detectors, the

	NA62 RICH	HIKE RICH
Sensor time resolution	240ps	100ps
Sensor quantum efficiency	20%	40%
Number of photo-electrons for π^+ at 15 GeV/c	7	14
Number of photo-electrons for π^+ at 45 GeV/c	12	24
Time resolution for π^+ at 15 GeV/c	90ps	27ps
Time resolution for π^+ at 45 GeV/c	70ps	20ps

Table 2: Comparisons of the expected performance of the NA62 and HIKE RICH detectors [2]

above expectations were confirmed by a Monte Carlo sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3).

For the three sensor sizes, a test production of 1M Ke3 was used to determine the preliminary performance of the detector and compare this to the current NA62 setup. The plots in Figure 18 show the light distribution on the PMT arrays for the three sensor sizes: 3 mm 18 (a,b), 6 mm 18 (c,d), and 9 mm 18 (e,f) compared to the illumination on the current NA62 PMT arrays 17. Due to the better placement of the square sensors, it is possible to see a more uniform illumination of the PMT arrays and better coverage of the light ring, especially in the Jura side. It is clear to see that either sensor size would be an improvement on the current NA62 setup.

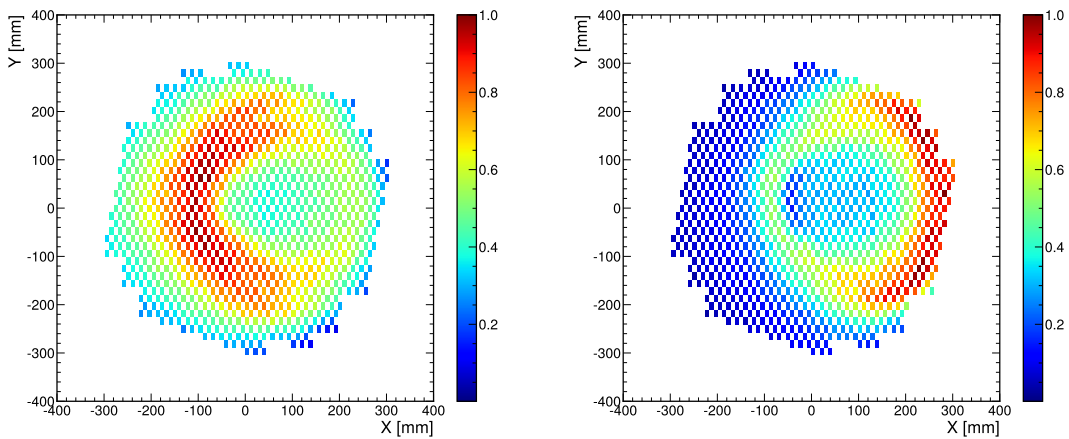
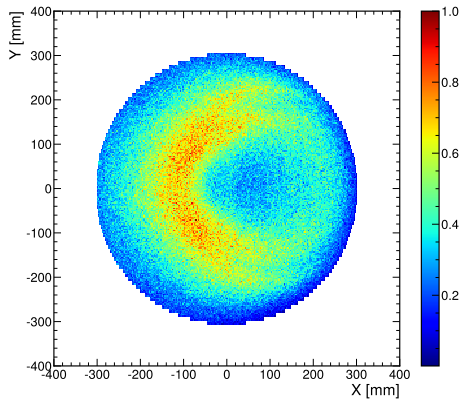
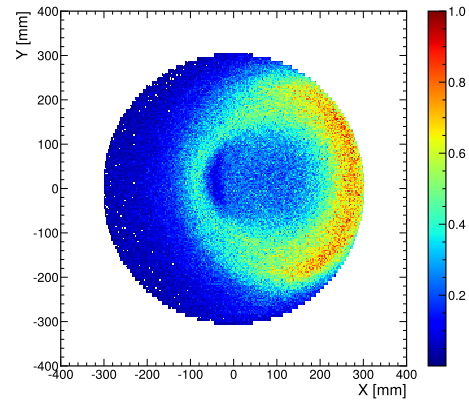


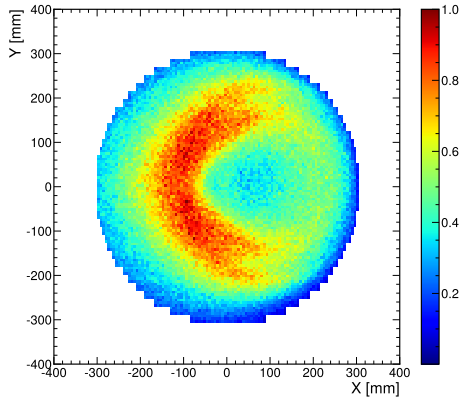
Figure 17: Cherenkov light illumination of RICH PMT arrays for the NA62 PMT arrays produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3). The left column shows illumination on the Saleve side of the RICH detector, and the right column shows the Jura side.



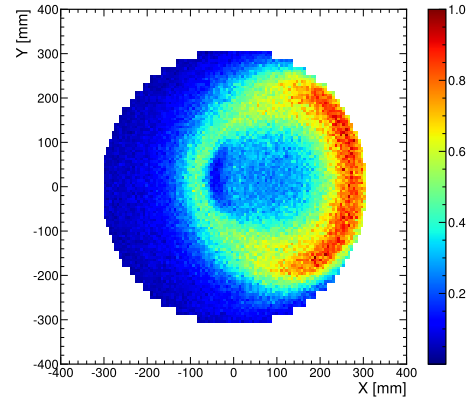
(a) 3 mm Saleve.



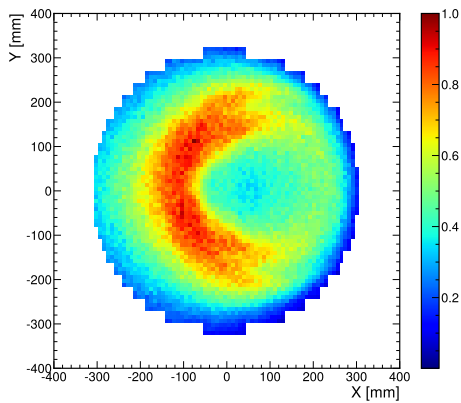
(b) 3 mm Jura.



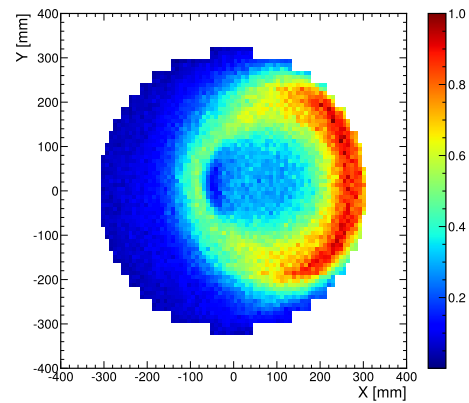
(c) 6 mm Saleve.



(d) 6 mm Jura.



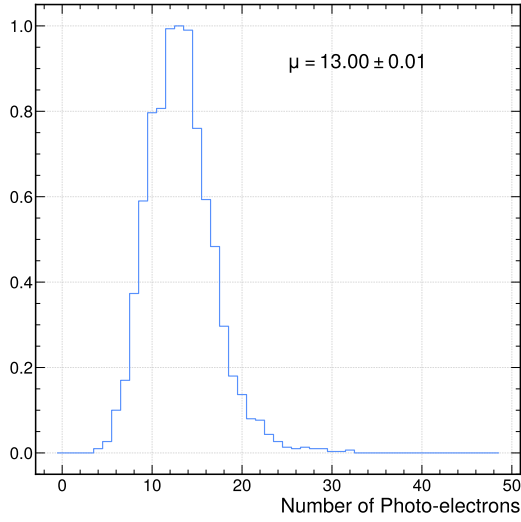
(e) 9 mm Saleve.



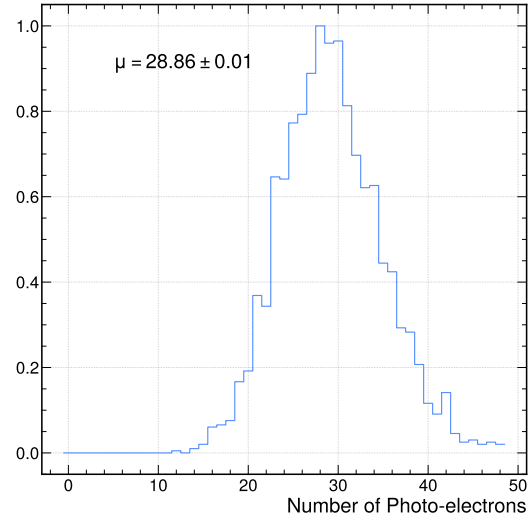
(f) 9 mm Jura.

Figure 18: Cherenkov light illumination of RICH PMT arrays for the three different sensors produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3). The left column shows illumination on the Saleve side of the RICH detector, and the right column shows the Jura side.

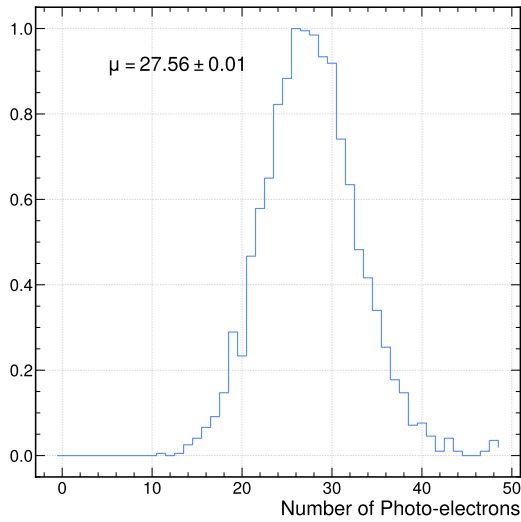
In the reconstruction, the photo-electrons are clustered into ring candidates. The number of reconstructed photo-electrons is shown in Figure 19, with the SiPM sensors resulting in a two-times increase in the number of hits per ring candidate. The improved number of reconstructed photo-electrons was mainly due to the enhanced quantum efficiency of the SiPMs and the sensors' better placement. This reduced the probability of multiple photons hitting the same sensor. There are minimal differences between the three sensor sizes shown in Figure 19 (b-d), with the 3 mm sensors having a slightly higher number of hits per ring candidate when compared to the 9 mm sensors.



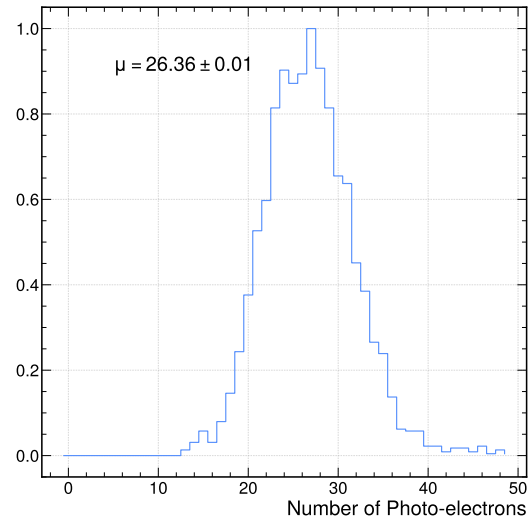
(a) NA62 PMT arrays.



(b) 3 mm SiPMs.



(c) 6 mm SiPMs.

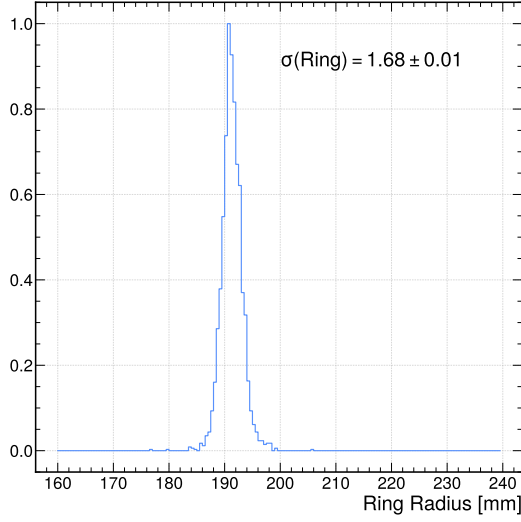


(d) 9 mm SiPMs.

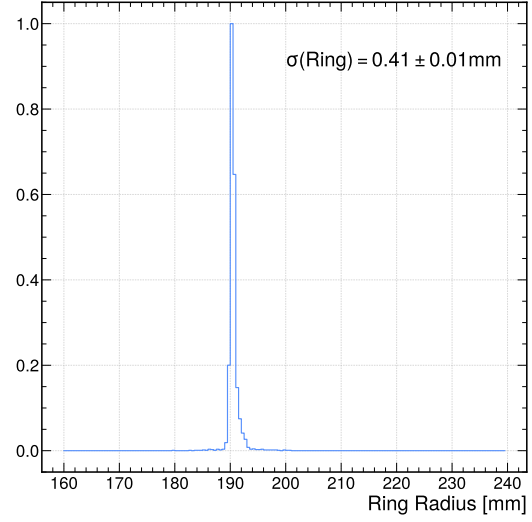
Figure 19: Number of photo-electrons per ring candidate for NA62 PMTs and the three possible HIKE SiPMs produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3).

The subsequent reconstructed ring candidate radius is shown in Figure 20, with the NA62 PMT array producing a resolution of 1.68 mm compared to 0.72 mm for the HIKE array with 9 mm SiPMs. With the the largest sensor size at 3 mm improving further to 0.41 mm. However, it would not be possible to obtain this resolution in reality due to small misalignments in the mirror and dispersion in the radiator gas. It was accepted that the 9 mm value would be the best possible resolution

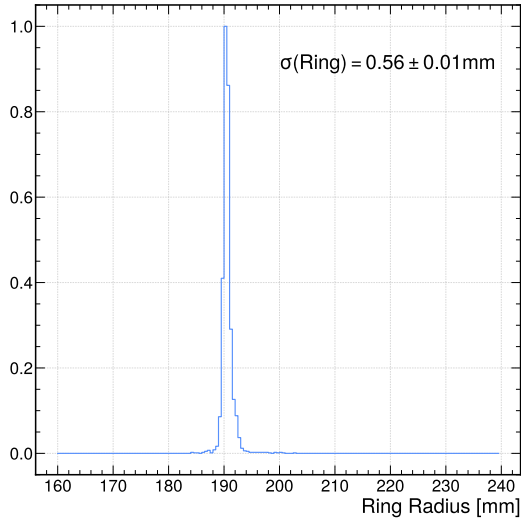
obtainable.



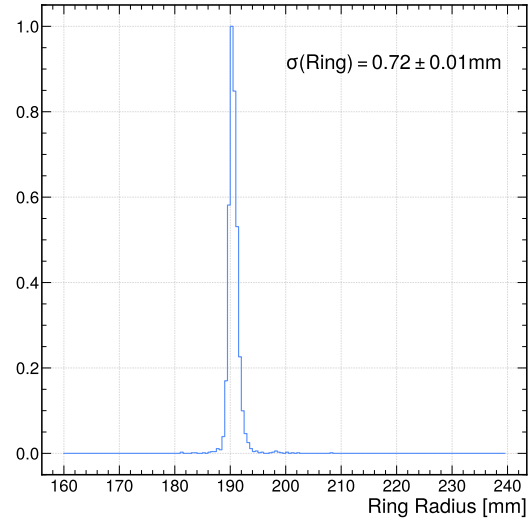
(a) NA62 PMT arrays.



(b) 3 mm SiPMs.



(c) 6 mm SiPMs.

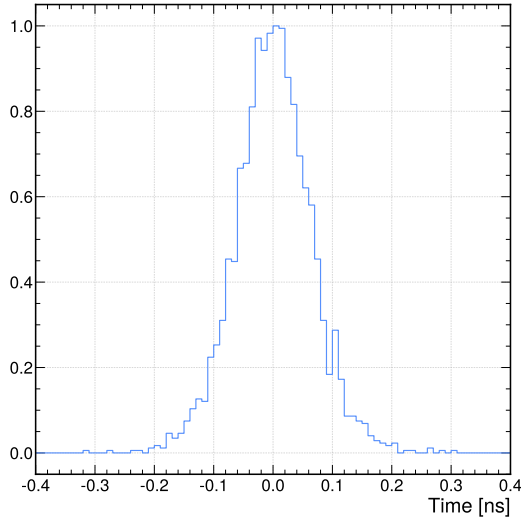


(d) 9 mm SiPMs.

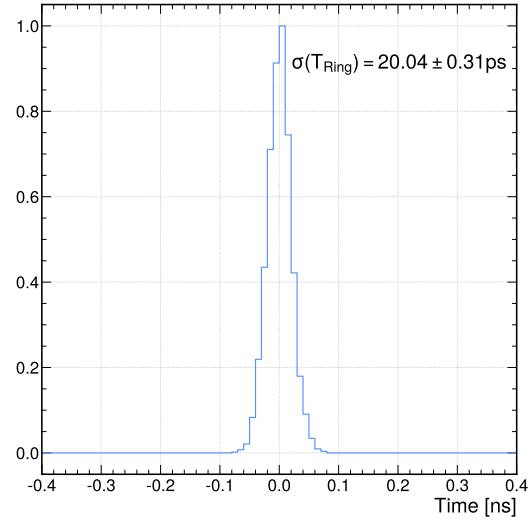
Figure 20: Reconstructed ring radius for electron candidates for NA62 PMTs and the three possible HIKE SiPMs produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3).

The time resolution of the RICH detector was calculated using the Ke3 sample by equally splitting the reconstructed photo-electrons of the ring candidates into two sets and calculating the time difference between them, with the results are shown in Figure 21. This method was used to remove any dependence on the reference time with respect to the single hit times. The 3x3 mm sensors produced the best

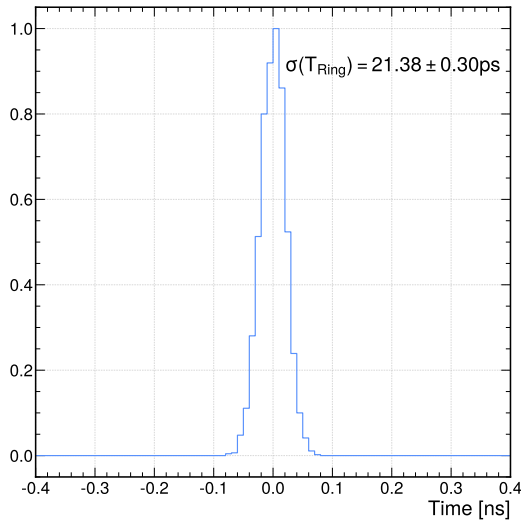
time resolution of 20.04 ps compared to the 9x9 mm sensors with a time resolution of 21.22 ps. However, there was little difference between the SiPMs, with each sensor size producing a time resolution of less than 30 ps, fulfilling the requirements of HIKE. The improvement was clear with using the small SiPMs when compared with the NA62 resolution in Figure 21, which had a time resolution of 61.35 ps.



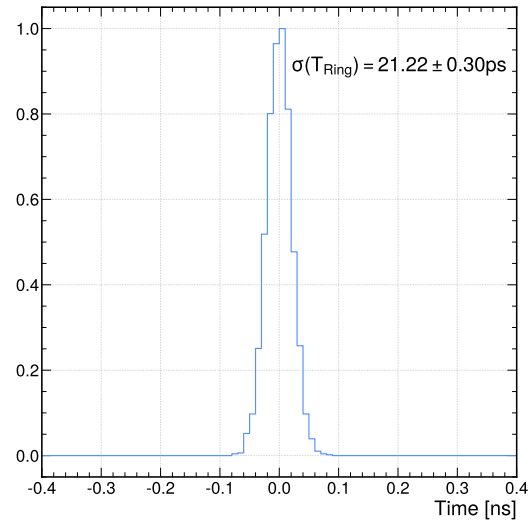
(a) NA62 PMT arrays.



(b) 3 mm SiPMs.



(c) 6 mm SiPMs.



(d) 9 mm SiPMs.

Figure 21: Time resolution for NA62 PMTs and the three possible HIKE SiPMs, calculated from the time difference between two equally populated subsets of hits within the ring candidate. A detailed description of the procedure is provided here [17]

After the RICH MC study, it was proven that introducing small SiPMs would dramatically increase the granularity of the detector. All three SiPM sensor sizes produced results which exceeded the requirements for HIKE with a time resolution of around 20 ps and improvements on the reconstructed ring radius resolution.

From the results of the HIKE simulation, the 9x9 mm SiPMs would be used as they provided excellent performance and would reduce the cost and complexity of implementing the smaller sensor sizes in the detector when it was built. A clear benefit was reducing the number of channels from 72780 for the 3x3 mm compared to 8588 for the 9x9 mm SiPMs.

0.4 Detector Rates

After the new GTK and RICH detectors were developed, it was necessary to determine the expected particle rate in the upgraded and non-upgraded detectors. The upgraded STRAW detector utilising 5 mm diameter straws was also included in the HIKE simulations. It was possible to estimate the expected rate for Phase 1 by scaling the known NA62 rates by the expected increase in beam intensity by a factor of 4 compared to NA62. The rates for Phase 2 were calculated using the new simulation software described in the previous sections.

An array of input files were produced in BDSIM [18] by the CERN beam group which contained the simulated HIKE beam components at a Z of 120 m, which was the start of the fiducial volume. The files were then used as an input for the NA62MC software and used as the primary beam instead of the turtle beam, which is the default for NA62. Each file contained 10000 events, equaling 10000 protons on target, and there were 18K files produced in total, which was equivalent to 1.8×10^8 protons on target. As this wouldn't be enough to get a reasonable estimate of the rates, each file was used 11 times as an input to the simulation with random seeds to increase the statistics. The simulation was run with the unforced flag which allows for four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$. After the simulation, the files were then reconstructed using the NA62 Reconstruction software, and the rates were calculated by scaling the numbers in the output by the expected HIKE Phase 2 beam conditions, a spill time of 4.8 seconds and a beam intensity of 2×10^{13} protons per spill.

The K_L rate at the entrance of the fiducial volume was calculated to be 304.1 MHz. The rate of the four main K_L decays inside the fiducial volume is shown in Table 3.

KL decay	Rate [MHz]
$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0)	13.89
$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$)	9.63
$K_L \rightarrow \pi^0 \pi^0 \pi^0$	6.60
$K_L \rightarrow \pi^+ \pi^- \pi^0$	4.45

Table 3: Rates of the four main K_L decays in the HIKE phase 2 fiducial volume.

The total rates from all particles in the events where the K_L decays are shown in Table 4. To calculate the rate, all hits in each detector's channels were collected and converted to the expected rate. The rates were large, but this was to be expected with a high intensity. However, the rate in the spectrometer for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay was concerning as one would expect a much lower rate compared to the other decays due to the lack of a charged daughter particles. Even with a lower branching ratio, it is expected that the $K_L \rightarrow \pi^+ \pi^- \pi^0$ decay should have a higher rate in the spectrometer due to the charged pions.

Detector	K_L Decay [MHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	126.8	79.9	50.5	49.2
STRAW-2	162.8	85.1	80.7	58.0
STRAW-3	165.7	89.3	67.7	56.1
STRAW-4	189.2	93.6	87.5	64.4
NCHOD	10.3	5.2	4.9	3.7
CHOD	90.5	33.4	61.3	33.8
IRC	3.8	1.2	4.2	2.0
LKr	609.5	266.0	440.7	298.3
MUV1	129.9	102.4	0.6	64.1
MUV2	10.3	17.9	0.1	4.7
MUV3	0.9	5.2	0.2	0.6
HAC	10.0	11.5	0.3	6.1

Table 4: Total rates in the detectors for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$.

Directly looking at the MC, it was possible to find the origin of the particles that produced the hits in the spectrometer. The production Z compared against the production radius of particles which produced a hit in the Spectrometer is shown in Figure 22. It is clear to see the four the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay, the majority (80 %) of the particles were produced after Z=210 m with most produced in the walls of the vacuum decay tube and the spectrometer chambers themselves, with only 20 % of particles produced before Z=210 m which would be consistent with being produced in the decay.

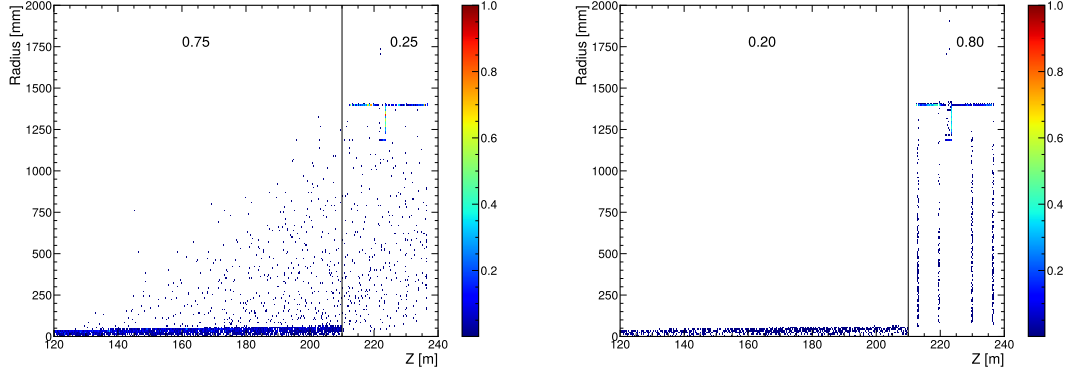


Figure 22: Production Z compared against production radius of particles producing a hit in the Spectrometer with the fraction of particles produced before and after $Z=210$ m. Left, for the $K_L \rightarrow \pi^+\pi^-\pi^0$ decay and right, for the $K_L \rightarrow \pi^0\pi^0\pi^0$ decay.

This is clear in Figure 23 where the MC particle identification code of the particles producing hits in the spectrometer is shown. For the $K_L \rightarrow \pi^0\pi^0\pi^0$ the majority of the particles are electron/positron pairs, produced in interactions of photons in the walls of the vacuum decay tube and in the spectrometer chambers, compared to the $K_L \rightarrow \pi^+\pi^-\pi^0$ decay where the majority of the particles are pions produced in the decay.

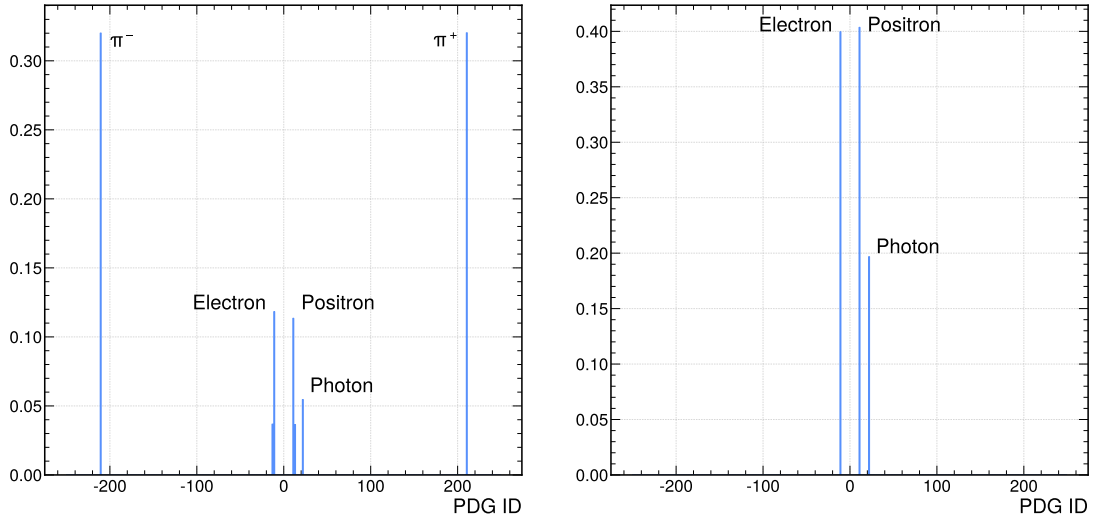


Figure 23: Monte Carlo particle code for the particles producing a hit in the Spectrometer. Left, for the $K_L \rightarrow \pi^+\pi^-\pi^0$ decay and right, for the $K_L \rightarrow \pi^0\pi^0\pi^0$ decay.

As well as the overall hit rate, it is important to understand the rate in the

detectors produced directly from the daughter particles of the K_L decays. The daughter particles were selected as particles with a GEANT4 production process of "Decay", which should include any subsequent decays of the daughter particles, e.g. $\pi^0 \rightarrow \gamma\gamma$. This should exclude all hits produced from particles produced by "gamma conversion", which produce electron-positron pairs and impact the rate, as described above. The rates for each detector are shown in Table 5. It was clear that the overall rates were much lower when compared to the total rates in Table 4 and that the rates in the spectrometer for the $K_L \rightarrow \pi^0\pi^0\pi^0$ decay are much lower which is expected. Overall, the rates were acceptable for the detectors as it was assumed that most of the hits produced not from the decay of the K_L or other subsequent decays would be removed by the trigger system as they would be outside the time window of the K_L decay.

Detector	K_L Decay Particle Rates [MHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0\pi^0\pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+\pi^-\pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	14.7	11.2	1.3	5.6
STRAW-2	15.4	11.7	2.1	6.2
STRAW-3	16.1	12.1	2.5	6.6
STRAW-4	16.5	12.2	3.4	6.9
NCHOD	19.9	10.8	9.4	7.0
CHOD	16.1	11.3	8.1	7.8
IRC	1.9	0.8	1.8	1.0
LKr	15.4	11.6	19.9	10.5
MUV1	7.6	9.8	0.3	4.7
MUV2	2.7	7.2	0.1	1.5
MUV3	1.1	6.7	0.3	0.9
HAC	0.6	1.0	0.1	0.3

Table 5: Detector rates for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$), $K_L \rightarrow \pi^0\pi^0\pi^0$ and $K_L \rightarrow \pi^+\pi^-\pi^0$, produced directly from the daughter particles of the K_L decays.

It is important to understand the rates of the detectors in order to determine the

overall detector design, especially for the electronics that would need to be designed to meet the high rates. The reported rates would also drive the development of the overall experiment data acquisition and trigger system. Along with the overall rate, it was vital to know the max channel rates in the detectors to design the electronics to handle the highest rate expected. The max channel rates for each detector were collected and are shown in Table 6.

Detector	K_L Decay Daughter Max Channel Rates [kHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	325	254	51	186
STRAW-2	377	254	27	154
STRAW-3	329	265	19	138
STRAW-4	365	258	39	138
NCHOD	484	329	186	206
CHOD	1290	635	786	559
IRC	1020	345	1063	516
LKr	575	321	385	293
MUV1	1830	1496	19	1032
MUV2	329	583	15	190
MUV3	79	234	39	43
HAC	206	210	27	138

Table 6: Max channel rates for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

As detailed previously in this section, the detectors which had designs finalised were implemented in the simulation, which allowed for correct rate calculation for all three of these detectors. However, other detectors would be upgraded to increase the granularity, but the design was not yet finalised and, therefore, not implemented in the simulation. This would result in an incorrect calculation as the rate is determined by the NA62 detector design and the layout of the channels in the detectors. As the final designs for these detector were not finalised, it was impractical to implement the

new detectors in the simulation. However, it was still possible to calculate the rates at the planes of the upgraded detectors by collecting particles passing through the Z of the detector. This was possible using raw MC, where information for particles is stored at different parts of the HIKE setup (checkpoints). The rates for charged particles produced from the four K_L decays at each checkpoint are shown in Table 7.

Checkpoints (Z Position)	K_L Decay Particle Rates [kHz/ cm^2]							
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 ± 0.11)%		$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 ± 0.07)%		$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 ± 0.12)%		$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 ± 0.05)%	
	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	19.28	0.34	16.64	0.25	0.05	10^{-3}	10.55	0.13
STRAW-3 Entry (230.5 m)	6.93	0.29	6.23	0.23	0.14	0.02	4.04	0.12
NCHOD Entry (237.6 m)	4.93	0.16	4.43	0.13	0.17	0.01	2.66	0.06
CHOD Entry (238.9 m)	4.60	0.27	4.07	0.21	0.14	0.02	2.55	0.11
IRC Entry (239.5 m)	4.77	0.26	3.89	0.21	0.13	0.02	2.43	0.11
LKr Entry (240.4 m)	4.60	0.15	3.72	0.13	0.10	0.01	2.34	0.06
MUV1 Entry (243.2 m)	3.92	0.02	3.38	0.08	0.03	10^{-4}	1.96	0.02
MUV2 Entry (244.4 m)	3.70	0.01	3.25	0.08	0.03	10^{-4}	1.82	0.01
MUV3 Entry (245.3 m)	0.15	0.08	1.05	0.08	0.01	10^{-7}	0.09	0.01
HAC Entry (253.3 m)	3.40	0.09	2.88	0.09	0.02	10^{-5}	1.44	0.01

Table 7: Particle rates at specified checkpoints along the HIKE detector setup. Including the average rate across the cross-section and the maximum rate per cm^2 for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

However, as the rates are collected from the cross-section at the expected detector position, the rates are higher than what would be seen due to the fact that all the detectors would eventually have a beam hole to accommodate beam particles passing through the detector. An example cross-section at the entry of the first STARW chamber is shown in Figure 24, where it is clear that the highest rate from the decay particles was found at the centre. As the detectors were not designed and the beam hole size was unknown, it was possible to estimate the size by calculating the beam divergence at the detector position. This was calculated using Equation 0.1.

$$\delta_{beam} = 0.4 \text{ mrad} \times Z \quad (0.1)$$

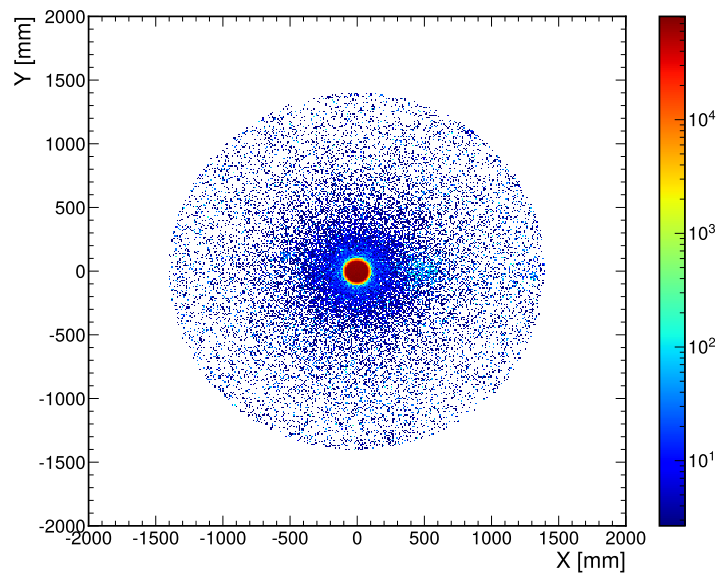


Figure 24: Rate of particles across the cross section at the first STRAW chamber entrance at $Z=213.$ m.

Using Equation 0.1, a mask for where the beam hole would be was applied to remove any particles that would not impact the detector. The resultant rates with the beam hole mask applied are shown in Table 8. As expected, this reduced the max rate for most detectors without impacting the average rate across the detector plane.

		K_L Decay Particle Rates [kHz/cm ²]							
		$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 ± 0.11)%		$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 ± 0.07)%		$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 ± 0.12)%		$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 ± 0.05)%	
Checkpoints (Z Position)	Beam radius [mm]	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	85.4	11.59	0.305	9.504	0.226	0.047	0.001	5.863	0.109
STRAW-3 Entry (230.5 m)	92.2	5.515	0.280	4.900	0.219	0.139	0.017	3.087	0.113
NCHOD Entry (237.6 m)	95.0	3.959	0.152	3.143	0.124	0.166	0.012	2.032	0.063
CHOD Entry (238.9 m)	95.6	3.418	0.256	3.017	0.202	0.139	0.017	1.849	0.104
IRC Entry (239.5 m)	95.8	3.418	0.249	2.921	0.198	0.133	0.016	1.733	0.100
LKr Entry (240.4 m)	96.2	2.501	0.145	2.053	0.122	0.100	0.010	1.314	0.059
MUV1 Entry (243.2 m)	97.3	0.279	0.016	1.311	0.074	0.033	10 ⁻⁴	0.227	0.012
MUV2 Entry (244.4 m)	97.8	0.167	0.008	1.215	0.069	0.027	10 ⁻⁴	0.116	0.007
MUV3 Entry (245.3 m)	98.1	0.153	0.008	1.051	0.079	0.007	10 ⁻⁷	0.089	0.007
HAC Entry (253.3 m)	101.3	0.167	0.008	1.147	0.078	0.013	10 ⁻⁶	0.138	0.007

Table 8: Particle rates at specified checkpoints along the HIKE detector setup excluding particles passing within the expected beam radius for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

For the detector design, it was also important to calculate the total rates produced by different particles, which is shown in Table 9. It's clear that the majority of the rates at the detector planes were from photons, especially in the spectrometer. However, the rate from charged particles, which the detectors would be mainly designed for, is reasonable, suggesting that the detectors could be designed and built with current technologies.

With all rates information from the simulation shown in this section, it would then be possible to design the detectors that were not already upgraded within the simulation. Table 9 was used in the HIKE proposal to demonstrate the requirement for upgrading the detectors to handle the increased rates. The particle rates are collected not from the geometrical acceptance of the corresponding sub-detectors but from the cross-section of the outer volume for the sub-detector found in Monte Carlo. For the Straw-1&3, a circle of radius of 141.5 mm, NCHOD a square of length 330 mm, CHOD a square of length 244 mm, IRC a square of length 122 mm, MEC a circle of radius 182.5 mm, MUV-1&2 a square of length 320 mm, MUV-3 a square of length 281 mm and HAC a square of length 281 mm. A geometrical cut has been made to remove particles that pass through the checkpoints within the expected

0.5 Conclusions

beam dispersion radius calculated to be $0.4 \times 10^{-3} \times Z$, where Z is the position of the checkpoint.

Checkpoints	Total Particle Rates kHz/cm ²									
	$e^{+/-}$		$\mu^{+/-}$		$\pi^{+/-}$		γ		Charged Particles	
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	264	0.26	224	1.58	117	0.41	13426	13.87	360	2.56
STRAW-3 Entry (230.5 m)	246	0.59	179	1.34	120	0.40	11076	14.02	291	2.62
NCHOD Entry (237.6 m)	216	0.45	174	1.13	115	0.24	11770	8.50	238	1.98
CHOD Entry (238.9 m)	318	2.96	176	1.18	112	0.41	11687	15.57	376	5.79
IRC Entry (239.5 m)	251	3.85	176	1.17	85	0.40	11481	15.58	291	5.64
MEC Entry (240.4 m)	85	1.76	174	1.12	109	0.24	166	0.30	238	3.25
MUV1 Entry (243.2 m)	5	0.01	190	1.04	32	0.03	29	0.01	190	1.07
MUV2 Entry (244.4 m)	8	10^{-3}	162	1.00	5	10^{-3}	5	10^{-3}	163	0.98
MUV3 Entry (245.3 m)	5	10^{-4}	115	0.89	3	10^{-4}	3	10^{-3}	115	0.88
HAC Entry (253.3 m)	11	0.02	120	0.89	5	10^{-3}	3	10^{-3}	120	0.89

Table 9: Average and Max particle rates per cm^2 for specified checkpoints along the length of the detector.

0.5 Conclusions

Unfortunately, in 2024, CERN management decided that HIKE would not be accepted for operation in ECN3. However, the development of the detectors, software and the subsequent physics studies was utilised for the KOTO II proposal [19].

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